

Lagrange Multiplier :-

$$\underline{f} + \alpha \underline{g} = 0$$

let $z = f(x, y)$ for maxima or minima

$$dz = 0$$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

where dx & dy are independent & not zero, it follows that extremum exist, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$

now let $g(x, y) = 0$ be a curve in xy plane. let us find points on the curve $g=0$ at which f is stationary

$$\therefore g=0 \Rightarrow dg=0$$

$$\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy = 0$$

$$\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y} = -\frac{\partial g / \partial x}{\partial g / \partial y}$$

$$\frac{\partial f / \partial x}{\partial g / \partial x} = \frac{\partial f / \partial y}{\partial g / \partial y} = -\alpha$$

$$\frac{\partial f}{\partial x} + \alpha \frac{\partial g}{\partial x} = 0 ; \quad \frac{\partial f}{\partial y} + \alpha \frac{\partial g}{\partial y} = 0$$

But these, together with constraint $g(x, y) = 0$ are the condition required for the function

$$h(x, y, \alpha) = f(x, y) + \alpha g(x, y)$$

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial x} + \alpha \frac{\partial g}{\partial x} = 0$$

$$\frac{\partial h}{\partial y} = \frac{\partial f}{\partial y} + \alpha \frac{\partial g}{\partial y} = 0$$

$$\& \frac{\partial h}{\partial \alpha} = g = 0$$

$$\left\{ \begin{array}{l} \Delta u_{\text{simple}} = \text{cte} \\ \frac{M_L}{M_i} = \text{cte} \\ \left. \begin{array}{l} \varepsilon_i \\ I_{sp_i} \end{array} \right\} = \text{cte} \end{array} \right\} \rightarrow \begin{array}{l} f_i \text{ and } \lambda_i \\ \text{that optimize} \\ \left(\frac{M_L}{M_i} \right) \end{array}$$

Optimization of multistage Rocket :-

$$\Delta u_{b,imp} = \sum_{i=1}^n g_e I_{sp,i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right)$$

fixed for maximization of $\frac{M_L}{M_{01}}$ known Fixed fixed

optimize λ_i

$$\lambda_i = \frac{M_{Li}}{M_0 - M_{Li}}$$

Variation $\leftarrow \delta(\Delta u_{b,imp}) = 0$

$$\delta \left[\sum_{i=1}^n g_e I_{sp,i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right) \right] = 0$$

$$\Rightarrow \sum_{i=1}^n \delta \left[g_e I_{sp,i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right) \right] = 0$$

$$\delta = \frac{\partial}{\partial \lambda_i} d\lambda_i$$

$$\Rightarrow \sum_{i=1}^n \frac{\partial}{\partial \lambda_i} \left[g_e I_{sp,i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right) \right] d\lambda_i = 0$$

$$\sum_{i=1}^n g_e I_{sp,i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)} d\lambda_i = 0$$

$$\therefore \sum_{i=1}^n g_e I_{sp,i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)} d\lambda_i = 0$$

$\rightarrow$ (A)

$$\delta(\Delta u_{b,imp}) = 0$$

$$\frac{\partial}{\partial \lambda_i} [(*)] d\lambda_i = 0$$

$$\frac{M_L}{M_{01}} = \frac{1}{\pi} \left(\frac{\lambda_i}{1+\lambda_i} \right)$$

$$\delta \left(\frac{M_L}{M_{01}} \right) = 0$$

Second constraint

$$\Rightarrow \delta \left[\frac{1}{\pi} \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

(or)

$$\sum_{i=1}^n \delta \left[\ln \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$\sum_{i=1}^n \frac{1}{\lambda_i (1+\lambda_i)} d\lambda_i = 0$$

$\rightarrow$ (B)

$$\delta \left(\frac{M_L}{M_{01}} \right) = 0$$

$$\sum_{i=1}^n Q_i + \sum_{i=1}^n \Delta_i = 0$$

$$[Q_1 + Q_2 + \dots + Q_n] + [\Delta_1 + \Delta_2 + \dots + \Delta_n] = 0$$

$$Q_i = 0$$

$$\Delta_i = 0$$

$$\Rightarrow Q_i + \Delta_i = 0$$

if we use Lagrange multiplier for (A) + (B) then

(B) + α (A) = 0 to get λ_i

$$\underbrace{\sum_{i=1}^n \frac{1}{\lambda_i (1+\lambda_i)} \delta \lambda_i}_B + \alpha \underbrace{\left(\sum_{i=1}^n g_e I_{SP_i} \frac{E_i - 1}{(1+\lambda_i)(E_i + \lambda_i)} \delta \lambda_i \right)}_A = 0$$

Here λ_i is independent

$$\frac{1}{\lambda_i (1+\lambda_i)} + \alpha g_e I_{SP_i} \frac{E_i - 1}{(1+\lambda_i)(E_i + \lambda_i)} = 0 \rightarrow (C)$$

$$\frac{1}{\lambda_i (1+\lambda_i)} = \frac{(1-E_i) \alpha g_e I_{SP_i}}{(1+\lambda_i)(E_i + \lambda_i)}$$

$$\frac{E_i + \lambda_i}{\lambda_i} = (1-E_i) \alpha I_{SP_i} g_e$$

$$\frac{E_i}{\lambda_i} + 1 = \alpha I_{SP_i} g_e (1-E_i)$$

$$\frac{E_i}{\lambda_i} = \underbrace{\alpha I_{SP_i} g_e (1-E_i)}_{\text{known}} - 1$$

$$\lambda_i = \frac{E_i}{\underbrace{\alpha g_e I_{SP_i} (1-E_i)}_{\text{unknown}} - 1} \rightarrow (D)$$

' α ' is obtained from the eq $\Delta U_{b,imp,tot}$

$$\Delta U_{b,imp,tot} = \sum_{i=1}^n g_e I_{SP_i} \ln \left(\frac{1+\lambda_i}{E_i + \lambda_i} \right)$$

$\downarrow$ fixed $\downarrow$ known $\downarrow$ fixed $\downarrow$ fixed

$$\Delta U_{b,imp} = \sum_{i=1}^n g_e I_{SP,i} \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}} \right]$$

λ_i

$$I = \sum_{i=1}^n \left[\frac{g_e I_{SP,i}}{\Delta U_{b,imp}} \right] \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}} \right]$$

λ_i

$$I = \sum \square \ln \square$$

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$$e^I = e^{\sum \ln \square \square}$$

$$I = \sum_{i=1}^n \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}} \right] \left[\frac{g_e I_{SP,i}}{\Delta U_{b,imp}} \right]$$

$$e = e^{\sum_{i=1}^n \ln \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}} \right] \left[\frac{g_e I_{SP,i}}{\Delta U_{b,imp}} \right]}$$

$$e^{\sum_{i=1}^n \ln \square \square} = \prod_{i=1}^n \square \square$$

$$e = \prod_{i=1}^n \left[\frac{1 + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}}{\epsilon_i + \frac{\epsilon_i}{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1}} \right]^{\frac{g_e I_{SP,i}}{\Delta U_{b,imp}}}$$

Simplified

$$e = \prod_{i=1}^n \left(\frac{\alpha g_e I_{SP,i} (1-\epsilon_i) - 1 + \epsilon_i}{\epsilon_i \alpha g_e I_{SP,i} (1-\epsilon_i) - \cancel{\epsilon_i} + \cancel{\epsilon_i}} \right)^{\frac{g_e I_{SP,i}}{\Delta U_{b,imp}}}$$

$$e = \prod_{i=1}^n \left(\frac{\cancel{\alpha g_e I_{SP,i} (1-\epsilon_i)} - 1 + \epsilon_i}{\epsilon_i \cancel{\alpha g_e I_{SP,i} (1-\epsilon_i)}} \right)^{\frac{g_e I_{SP,i}}{\Delta U_{b,imp}}}$$

$$\prod_{i=1}^n Q_i = Q_1 \times Q_2 \times Q_3 \times \dots \times Q_n$$

$$\sum_{i=1}^n Q_i = Q_1 + Q_2 + Q_3 + \dots + Q_n$$

$$= \prod_{i=1}^n \left[\frac{1}{\epsilon_i} - \frac{(1 - \epsilon_i)}{\epsilon_i \alpha g_{e, \text{ISP}_i} (1 - \epsilon_i)} \right]^{\frac{g_{e, \text{ISP}_1}}{\Delta b, \text{imp}}}$$

Neper No.
2.71 = e = $\prod_{i=1}^n \left[\frac{1}{\epsilon_i} - \frac{1}{\epsilon_i \alpha g_{e, \text{ISP}_i}} \right]^{\frac{g_{e, \text{ISP}_1}}{\Delta b, \text{imp}}}$

$$\therefore \left[\frac{1}{\epsilon_1} - \frac{1}{\epsilon_1 \alpha g_{e, \text{ISP}_1}} \right]^{\frac{g_{e, \text{ISP}_1}}{\Delta b, \text{imp}}} \times \left[\frac{1}{\epsilon_2} - \frac{1}{\epsilon_2 \alpha g_{e, \text{ISP}_2}} \right]^{\frac{g_{e, \text{ISP}_2}}{\Delta b, \text{imp}}} \dots \left[\frac{1}{\epsilon_n} - \frac{1}{\epsilon_n \alpha g_{e, \text{ISP}_n}} \right]^{\frac{g_{e, \text{ISP}_n}}{\Delta b, \text{imp}}} = C = 2.718$$

Solve this & get α
implicit, non-linear Eqⁿ
for $\alpha \rightarrow$ Use MATLAB

if $\alpha = \frac{2+n}{3}$ ^{known} : explicit, linear

Example

Reconsider the earlier three-stage rocket example:

$$\left. \begin{aligned} I_{sp1} &= I_{sp2} = I_{sp3} = 370 \text{ s} \\ \epsilon_1 &= 0.05, \epsilon_2 = 0.0698, \epsilon_3 = 0.1892 \end{aligned} \right\} \text{ given}$$

$$\frac{M_L}{M_{01}} = \frac{68,000}{2,500,000} = 0.0272$$

Computed

$$\text{With } \lambda_1 = 0.3021, \lambda_2 = 0.4464, \lambda_3 = 0.6126,$$

$$\Delta u_{b,imp} = 11023 \text{ m/s} \rightarrow \text{fixed} \rightarrow \delta[\Delta u_{b,imp}] = 0$$

Let us optimize the vehicle to maximize $\left(\frac{M_L}{M_{01}}\right)$

$$\text{for } \Delta u_{b,imp} = 11023 \text{ m/s}$$

$$\left\{ \frac{1}{\epsilon_1} \left[1 - \frac{1}{\alpha g_e I_{sp1}} \right] \right\}^{\frac{g_e I_{sp1}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_2} \left[1 - \frac{1}{\alpha g_e I_{sp2}} \right] \right\}^{\frac{g_e I_{sp2}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_3} \left[1 - \frac{1}{\alpha g_e I_{sp3}} \right] \right\}^{\frac{g_e I_{sp3}}{\Delta u_{b,imp}}} = e \quad \text{⑤}$$

Use MATLAB to solve Equation ⑤ for αg_e

Then, get λ_i from Eqn. ④:

$$\lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1}$$

$$\frac{M_L}{M_{01}} = \frac{\lambda_1}{1 + \lambda_1} \cdot \frac{\lambda_2}{1 + \lambda_2} \cdot \frac{\lambda_3}{1 + \lambda_3} \quad [\text{from (10.35)}]$$

Example

Consider a 3-stage rocket.

$$\left\{ \begin{array}{l} \epsilon_1 = 0.0765, \epsilon_2 = 0.114, \epsilon_3 = 0.111 \\ I_{sp1} = 304.5, I_{sp2} = 421.5, I_{sp3} = 421.5 \end{array} \right.$$

$$\Delta u_{b,imp} = 12.4 \text{ km/s}$$

Optimize the vehicle for maximum $\left(\frac{M_L}{M_{01}}\right)$.

$$\left\{ \frac{1}{\epsilon_1} \left[1 - \frac{1}{\alpha g_e I_{sp1}} \right] \right\}^{\frac{g_e I_{sp1}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_2} \left[1 - \frac{1}{\alpha g_e I_{sp2}} \right] \right\}^{\frac{g_e I_{sp2}}{\Delta u_{b,imp}}} \otimes \left\{ \frac{1}{\epsilon_3} \left[1 - \frac{1}{\alpha g_e I_{sp3}} \right] \right\}^{\frac{g_e I_{sp3}}{\Delta u_{b,imp}}} = e \quad \text{--- (E)}$$

Use MATLAB to solve Equation (E) for αg_e

Then, get λ_i from Eqn. (D):

$$\lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1}$$

$$\frac{M_L}{M_{01}} = \frac{\lambda_1}{1 + \lambda_1} \cdot \frac{\lambda_2}{1 + \lambda_2} \cdot \frac{\lambda_3}{1 + \lambda_3} \quad [\text{from (10.35)}]$$

MATLAB solution:

$$\alpha g_e = 0.0038966$$

$$\lambda_1 = 0.8142, \lambda_2 = 0.2514, \lambda_3 = 0.2421$$

$$\frac{M_L}{M_{01}} = 0.01758$$

MATLAB solution:

$$\alpha g_e = 0.003554$$

$$\lambda_1 = 0.2005, \lambda_2 = 0.3125, \lambda_3 = 2.8524$$

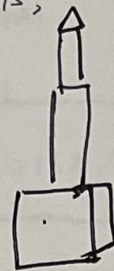
$$\left(\frac{M_L}{M_{01}} \right)_{opt} = 0.0294$$

$$I_{sp} g_e \ln \frac{1+\lambda_1}{\epsilon_1+\lambda_1}$$

$$\Delta u_{b, imp1} = 5688.1 \text{ m/s}, \Delta u_{b, imp2} = 4477.2 \text{ m/s},$$

$$\Delta u_{b, imp3} = 857.7 \text{ m/s}$$

$$\Delta u_{b, imp} = \sum_{i=1}^3 \Delta u_{b, imp i} = 11023 \text{ m/s}$$



* If M_L is maintained at 68000 kg,

$$(M_{01})_{opt} = \frac{M_L}{(M_L/M_{01})_{opt}} = \frac{68000}{0.0294} \approx \underline{2,313,000 \text{ kg}}$$

Thus the optimized design is capable of imparting the Δu_b of 11023 m/s to the 68000 kg payload with a lighter launch vehicle [2,313,000 kg vs 2,500,000 kg]

* If, on the other hand, M_{01} is maintained at 2,500,000 kg,

$$(M_L)_{opt} = M_{01} \left(\frac{M_L}{M_{01}} \right)_{opt} = 2,500,000 (0.0294) = \underline{73500 \text{ kg}}$$

Thus the optimized 2,500,000 kg launch vehicle is capable of providing the Δu_b of 11023 m/s to a more massive payload (73650 kg vs. 68000 kg)

$$\rightarrow I_{sp} g_0 \ln \frac{1+\lambda}{\epsilon+\lambda}$$

$$(\Delta u_{b, \text{imp}1} = 2127.5 \text{ m/s}, \Delta u_{b, \text{imp}2} = 5084.2 \text{ m/s},$$

$$\Delta u_{b, \text{imp}3} = 5194.3 \text{ m/s}$$

$$\Delta u_{b, \text{imp}} = \sum_{i=1}^3 \Delta u_{b, \text{imp}i} = 12400 \text{ m/s} \rightarrow \text{fixed}$$

The data of this problem correspond to the Saturn V lunar launch vehicle. Its initial mass (M_{01}) was 2,902,000 kg and the mass of the Apollo 11 payload (M_L) was 47000 kg, which gives $\frac{M_L}{M_{01}} = 0.0162$.

A vehicle with stages having equal ϵ_i and I_{sp_i} values is said to be similarly staged. It is clear from Eqn (D), that the stages will have equal values of λ_i as well. For such a vehicle, Eqn. (E)

becomes
$$\Delta u_{b, \text{imp}, \text{tot}} = \sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right)$$

Solve for λ

$$n g_e I_{sp} \ln \left(\frac{1+\lambda}{\epsilon+\lambda} \right) = \Delta u_{b, \text{imp}, \text{tot}} \quad (F)$$

$$\sum_1^3 2 = 3 \times 2$$

where n is the number of stages, and

$$I_{sp} \stackrel{\text{def}}{=} I_{sp1} = I_{sp2} = I_{sp3},$$

$$\lambda \stackrel{\text{def}}{=} \lambda_1 = \lambda_2 = \lambda_3, \text{ and}$$

$$\epsilon \stackrel{\text{def}}{=} \epsilon_1 = \epsilon_2 = \epsilon_3.$$

$$\sum_1^n \bigcirc = n \bigcirc$$

the same object

In this case, λ can be found from Eqn. (F), without

$$\frac{M_L}{M_{0,1}} = \prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right)$$

$$\prod_{i=1}^n \bigcirc = \bigcirc^n$$

having to find α . If needed, α can be found from Egn. (D) $\left\{ \lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1} \right\}$ once the same thing has been

determined.

Example

Reconsider the first three-stage example $\{M_{0,1} = 2,500,000 \text{ kg}, \Delta u_b = 11023 \text{ m/s}\}$. The stages all have the same I_{sp} . $M_{S_{tot}} = 145,000 \text{ kg}$, $M_{P_{tot}} = 2,287,000 \text{ kg}$. If the structural and propellant masses are redistributed such that $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon$, ^{def}

$$\epsilon = \frac{M_{S1}}{M_{S1} + M_{P1}} = \frac{M_{S2}}{M_{S2} + M_{P2}} = \frac{M_{S3}}{M_{S3} + M_{P3}}$$

$$\frac{M_{S_{tot}}}{(M_{S1} + M_{P1}) + (M_{S2} + M_{P2}) + (M_{S3} + M_{P3})} = \frac{M_{S_{tot}}}{M_{S_{tot}} + M_{P_{tot}}} = \epsilon$$

$$M_{S_{tot}} + M_{P_{tot}}$$

$$\frac{145000}{145000 + 2287000} = 0.05962$$

With this value of ϵ , (F) $\Rightarrow \lambda = 0.4772$ [$n=3$]

$$\frac{M_L}{M_{0,1}} = \left(\frac{\lambda}{1+\lambda} \right)^3 = 0.0337 \quad (\text{larger than the } 0.0294 \text{ value})$$

obtained with $\epsilon_1 = 0.05$, $\epsilon_2 = 0.0698$ and $\epsilon_3 = 0.1892$.

The value of αg_e (not really needed) can be found from (D) :

$$\alpha g_e = \frac{1}{I_{sp}(1-\epsilon)} \left(1 + \frac{\epsilon}{\lambda} \right) = 0.03233$$

The similarly-staged vehicle has the highest $\left(\frac{M_2}{M_{01}} \right)$.